

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Remainder and Factor Theorems

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The factor theorem

1. Which factor is common to $x^2 - 4$ and $x^3 - x^2 - 4x + 4$? (a) $(x + 1)$ (b) $(x - 1)$ (c) $(x - 2)$ (d) $(x - 4)$ **ICSE 2025, 1 mark**

SOLUTION

$$x^2 - 4 = (x - 2)(x + 2)$$

$$x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1) = (x - 1)(x - 2)(x + 2)$$

Of the options, $(x - 2)$ divides both.

Answer: (c)

2 Factorising a cubic polynomial

2. $(x - 2)$ is a factor of $2x^3 - 7x^2 + kx - 2$. Find k , and then factorise the polynomial completely. **ICSE 2025, 3 marks**

SOLUTION

By the factor theorem $f(2) = 0$: $16 - 28 + 2k - 2 = 0$, so $k = 7$.

Divide $2x^3 - 7x^2 + 7x - 2$ by $(x - 2)$: the quotient is $2x^2 - 3x + 1$.

$$2x^2 - 3x + 1 = (2x - 1)(x - 1)$$

Answer: $k = 7$; $2x^3 - 7x^2 + 7x - 2 = (x - 2)(2x - 1)(x - 1)$

3. $(x - 1)$ is a factor of $3x^3 + 8x^2 - 15x + k$. Find k , and then factorise the polynomial completely. **ICSE 2024, 3 marks**

SOLUTION

$f(1) = 0$: $3 + 8 - 15 + k = 0$, so $k = 4$.

Divide $3x^3 + 8x^2 - 15x + 4$ by $(x - 1)$: the quotient is $3x^2 + 11x - 4$.

$$3x^2 + 11x - 4 = 3x^2 + 12x - x - 4 = (3x - 1)(x + 4)$$

Answer: $k = 4$; $(x - 1)(3x - 1)(x + 4)$

3 The remainder theorem

4. What must be subtracted from $x^3 + x^2 - 2x + 1$ so that the result is exactly divisible by $(x - 3)$?
(a) -31 (b) -30 (c) 30 (d) 31 **ICSE 2024, 1 mark**

SOLUTION

The number to subtract is the remainder on division by $(x - 3)$, which is $f(3)$.

$$f(3) = 27 + 9 - 6 + 1 = 31$$

Answer: (d)

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